

SOC Home Lab Deployment & Integration Guide

A streamlined setup manual to build an integrated Security Operations Center (SOC) home lab on a single Windows 11 host using Oracle VirtualBox, host-based telemetry agents, and external threat intelligence APIs.

1. Network & Virtual Machine Baseline

Deploy three core virtual machines in VirtualBox using **Bridged Networking** attached to your primary physical Wi-Fi/Ethernet adapter to allow bidirectional L3 routing across components.

- **pfSense Firewall:** Edge firewall and network gateway. Set Promiscuous Mode to **Allow All** on the WAN interface.
- **Wazuh Server (SIEM / Manager):** All-in-one Wazuh architecture (Manager, Indexer, and Dashboard).
- **Kali Linux:** Simulation attacker platform set up with standard testing toolkits (Nmap, Hydra).

2. Endpoint & Telemetry Setup (Windows 11 Host)

A. Microsoft Sysmon (EDR Telemetry)

1. Download Microsoft Sysmon alongside an updated configuration file (e.g., **sysmonconfig.xml**).

Open PowerShell as Administrator and run the installer:

- `.\Sysmon64.exe -accepteula -i sysmonconfig.xml`

Confirm the service status:

- `Get-Service -Name Sysmon64`

B. Suricata NIDS (Network Inspection)

1. Install **Npcap** with "WinPcap API-compatible mode" enabled.
2. Install **Suricata** to default path (**C:\Program Files\Suricata**).

Identify your active network interface GUID via PowerShell:

- `Get-NetAdapter | Select-Object Name, InterfaceGuid`

Open **suricata.yaml** and configure the capture interface under the **pcap** block:

pcap:

```
- interface: \Device\NPF_{YOUR-INTERFACE-GUID}
checksum-checks: auto
```

Launch Suricata in live capture mode:

- `cd "C:\Program Files\Suricata"`
- `.\suricata.exe -c "suricata.yaml" --pcap`

C. Wazuh Agent Deployment

1. Download and install the Wazuh Windows Agent.
2. Edit `C:\Program Files (x86)\ossec-agent\ossec.conf` to declare your Wazuh Server IP address:

```
<client>

<server>

  <address>YOUR_WAZUH_SERVER_IP</address>

  <port>1514</port>

  <protocol>tcp</protocol>

</server>

</client>
```

Append log harvesting blocks for Sysmon and Suricata `eve.json`:

```
<!-- Sysmon Telemetry -->

<localfile>

  <location>Microsoft-Windows-Sysmon/Operational</location>

  <log_format>eventchannel</log_format>

</localfile>

<!-- Suricata Alerts -->

<localfile>

  <location>C:\Program Files\Suricata\log\eve.json</location>

  <log_format>json</log_format>

</localfile>
```

Start the agent service:

- `Restart-Service -Name WazuhSvc`

3. Perimeter Telemetry: pfSense Syslog Forwarding

1. Log in to the pfSense webConfigurator ([Status](#) > [System Logs](#) > [Settings](#)).
2. Enable **Send log messages to remote syslog server**.
3. Set the remote server to [YOUR_WAZUH_SERVER_IP:514](#) using the [UDP](#) protocol.
4. Enable logging for Firewall Events and Authentication processes.

On the Wazuh Server, ensure [wazuh-remoted](#) is listening on port 514:

- `sudo ss -ulnp | grep 514`

4. Threat Intelligence Integrations (VirusTotal, AbuseIPDB & AlienVault OTX)

Wazuh automatically enriches alerts and suspicious telemetry by querying external Cyber Threat Intelligence (CTI) APIs through the [wazuh-integratord](#) daemon.

A. Obtain API Keys

- **VirusTotal:** Register at [VirusTotal](#) and copy your free Public API Key (used for automated file hash inspection via FIM / Sysmon).
- **AbuseIPDB:** Create an account at [AbuseIPDB](#) and generate a Key under the API tab (used for IP reputation scoring).
- **AlienVault OTX:** Sign up at [AlienVault OTX](#) and retrieve your OTX Key (used for pulse and IOC correlation).

B. Configure [ossec.conf](#) on Wazuh Server

Access your Wazuh Server via SSH and edit `/var/ossec/etc/ossec.conf`. Add the integration blocks inside the main `<ossec_config>` section:

```
<!-- VirusTotal Hash Lookup Integration -->
```

```
<integration>
```

```
<name>virustotal</name>
```

```
<api_key>YOUR_VIRUSTOTAL_API_KEY</api_key>
```

```
<group>syscheck,sysmon_event1</group>
```

```
<alert_format>json</alert_format>
```

```
</integration>
```

```
<!-- AbuseIPDB IP Reputation Integration -->
```

```
<integration>
```

```
<name>custom-abuseipdb</name>
```

```

<hook_url>https://api.abuseipdb.com/api/v2/check</hook_url>

<api_key>YOUR_ABUSEIPDB_API_KEY</api_key>

<rule_id>100114,100115</rule_id>

<alert_format>json</alert_format>

</integration>

```

```

<!-- AlienVault OTX Threat Intelligence Integration -->

<integration>

  <name>alienvault-otx</name>

  <api_key>YOUR_ALIENVAULT_OTX_API_KEY</api_key>

  <group>firewall_drop,authentication_failed</group>

  <alert_format>json</alert_format>

</integration>

```

C. Apply Changes & Validate Integration Service

Restart the Wazuh Manager to apply the new API hooks:

- `sudo systemctl restart wazuh-manager`

Verify that the integrator daemon spawned correctly:

- `sudo /var/ossec/bin/wazuh-control status | grep integratord`

Expected output: wazuh-integratord is running...

Monitor integration requests and API rate-limit logs in real time:

- `sudo tail -f /var/ossec/logs/integrations.log`